

Klocwork – Utilities Overview – V5.2

PCA Team
PD-Cybersecurity
SEP 2024



Visteon®

Agenda

- Overview
- Klocwork Report Extraction
- Klocwork Build Summary
- Klocwork Ignore files trim Build spec
- Klocwork trim the build spec for the desktop analysis
- Klocwork AutoSconf – Identify huge issues
- Klocwork desktop analysis report conversion JSON/XML to CSV
- Klocwork desktop analysis report conversion XML to HTML

1. **kwcsvexport** → The report extraction script helps to extract the issues report from the respective stream in the Klocwork server in CSV format based on the view.
2. **kwbuildsummary** → The build summary script is used to publish the Translation Units Coverage Summary report based on the Klocwork build\analysis execution.
3. **kwtrimbuildspec** → The kwtrimbuildspec script is used to ignore the 3rd party and unintended files\components which will be skipped to analyze and load in server.
4. **kwAutoSconf** → The kwAutoSconf is used to identify the defects from the build.log file which will be located inside kw_tables and create sconf file for the detected files with defects.
5. **SingleFileBuildspec** → This script is designed to trim the klocwork build spec compilation lines from a build specification based on path patterns provided, specifically for the Klocwork differential analysis.
6. **jsontocsv** → Once the Klocwork desktop analysis is done we can be able to generate the XML and the JSON results, This script is used to convert the json/xml data into csv data.
7. **Xmltohtml** → Once the Klocwork desktop analysis is done we can be able to generate the XML and the JSON results, This script is used to convert the xml data into html data.

1. Klocwork Report extraction

- The report extraction script helps to extract the issues report from the respective stream in the Klocwork server in CSV format based on the view.
 - By default, the script will extract the “**Open-Outstanding**” issues based on the input view for the respective stream from the Klocwork server.
 - Using **--total** option, the total issues report will be extracted from the server (**Note: Total issues Extraction will take extra time based on the issues count**)
 - Using **--hc** option, the highest category column will provide the promoted category for the respective taxonomy
 - Eg: For CERT L2, L3 will display together in the “**Category**” Column, if the team wants to have a single category, then this option can be used. Note that, this will promote the category to “L2” in this case
 - If user needs this info based on project need, then need to pass **-hc** as an argument to the utility.
 - Using the **--ccm** option, the cyclometric report we can extract from the server.
 - Using the **--loc** option, the file-based report we can extract from the server.
 - Using the **--tags** option, we can get the tags column, In that column will store the tag data which is mentioned in the modules in the Klocwork server project
- **Syntax:** `kwcsvexport -url <Klocwork url> --project/stream <Specify the stream name along with project name> --view "view name" --output-file <outputfilename>.csv`

Example Commands to generate a MISRAC report (default):

```
kwcsvexport.exe -url https://kw.group4.Visteon.com:8443/ --project/streamTYO_DI_VIP/TYO_DI_38XD_380D_VIP_MY24_EP29284  
--view "MISRAC-ISSUES" --output-file TYO_DI_38XD_380D_VIP_MY24_EP29284 _MISRA.csv
```

Klocwork Report Extraction – Sample CSV Report

```
C:\Klocwork_share\Script\new\Make_report_July_release\Sep_Release\kwcsvexport\Windows>kwcs
vexport --url https://kw.group2.visteon.com:8443/ --project/stream VW-SKODA_IVI_AOSP/VW-SK
ODA_IVI_MIB_MY24_EP30997 --view MISRAC-ISSUES --output-file EP30997_MISRA.csv
Current Version = 4.6
```

Project	Build	Build Date	View	Issue Count														
NIS_DI_VI	build_288	Tue Apr 02	MISRAC-IS	35,486														
Note:	For Staic refer Severity column "K", For MISRA/CERT refer Reference and Category column "P"																	
ID	DateOrigin	Filename	Method	Code	Message	Status	State	Owner	Severity	Severity	Url	Taxonomy	Reference	Line	Category	HighestCat	Modules	Comment
1101	Mar-07-20	H:\di.gen.r	Rte_Mdl_L	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	87	adv.	adv.	BSW_CFG	comment
1108	Mar-07-20	H:\di.gen.r	Rte_Mdl_L	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	112	adv.	adv.	BSW_CFG	comment
1109	Mar-07-20	H:\di.gen.r	Rte_Mdl_L	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	108	adv.	adv.	BSW_CFG	comment
1110	Mar-07-20	H:\di.gen.r	Rte_Mdl_L	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	105	adv.	adv.	BSW_CFG	comment
1164	Mar-07-20	H:\di.mod	SpeedtoFr	INVARIAN	Expression	Analyze	Existing	PINDHU	3	Warning	https://kw	MISRA C 2	14.3 (C99-	582	req.	req.	Nissan_SW	comment
1768	Mar-07-20	H:\di.mod	Mdl_DigiS	NUM.OVE	The expres	Analyze	Existing	PINDHU	3	Warning	https://kw	MISRA C 2	Dir. 4.1 (C9	358	req.	req.	Nissan_SW	comment
7356	Mar-07-20	H:\di.com	App_spec	NUM.OVE	The expres	Analyze	Existing	PINDHU	3	Warning	https://kw	MISRA C 2	Dir. 4.1 (C9	10323	req.	req.	Nissan_Mc	comment
8592	Mar-07-20	H:\di.mod	Mdl_RearS	MISRA.ET	An express	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	10.3 (C99-	6141	req.	req.	Nissan_SW	comment
8593	Mar-07-20	H:\di.mod	Mdl_RearS	MISRA.ET	An express	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	10.3 (C99-	7690	req.	req.	Nissan_SW	comment
8594	Mar-07-20	H:\di.mod	Mdl_RearS	MISRA.ET	An express	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	10.3 (C99-	7737	req.	req.	Nissan_SW	comment
8793	Mar-07-20	H:\di.mod	Md_enter	MISRA.ET	Operand to	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	10.1 (C99-	676	req.	req.	Nissan_SW	comment
24469	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	66	adv.	adv.	BSW_CFG	comment
24472	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	90	adv.	adv.	BSW_CFG	comment
24473	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	100	adv.	adv.	BSW_CFG	comment
24474	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	161	adv.	adv.	BSW_CFG	comment
24475	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	164	adv.	adv.	BSW_CFG	comment
24476	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	147	adv.	adv.	BSW_CFG	comment
24477	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	150	adv.	adv.	BSW_CFG	comment
24480	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	154	adv.	adv.	BSW_CFG	comment
24481	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	157	adv.	adv.	BSW_CFG	comment
24484	Mar-07-20	H:\di.gen.r	Rte_Mdl_F	MISRA.INC	Include dir	Analyze	Existing	PINDHU	4	Review	https://kw	MISRA C 2	20.1 (C99-	168	adv.	adv.	BSW_CFG	comment

2.Klocwork Build Summary

- The build summary script is used to publish the Translation Units Coverage Summary report based on the Klocwork build\analysis execution.
- In the result check the source file compiled and analyzed percentage and count.
- Inform the PCA team if there is a reduction in the percentage and the source file count.
- **Syntax:** `kwbuildsummary <Klocwork_mytables>`

Example Commands to generate a summary report(Windows):

`kwbuildsummary VP_Tables/`

```
psekar2@INDGXY7RQ3:~/Script_release/Sep_2024/kwbuildsummary/Linux$ ./kwbuildsummary RE-12/
Klocwork Build.log Summary Report
-----
Klocwork Version: 23.4.0.70
Klocwork Server: chipd098([172.17.0.1,10.218.228.82])
Klocwork Project: RE_IVI_AOSP-A12
Klocwork Tables Directory: /data/rebuild/RE-Workspace/qssi-12/RE_A12_my_tables
Build Date: July 3, 2024 3:16:43 PM UTC
Engine(s) used: CLASSIC, MODERN
Number of C/C++ source files compiled: 3651
Number of Java source files compiled: 12170
Number of C/C++ source files analyzed: 3651
Percentage of compiled C/C++ files analyzed: 100.00%
Number of C/C++ source files analyzed cleanly: 3650
Percentage of C/C++ files analyzed cleanly: 99.97%
Percentage of compiled C/C++ files analyzed cleanly: 99.97%
psekar2@INDGXY7RQ3:~/Script_release/Sep_2024/kwbuildsummary/Linux$
```

3.Klocwork Ignore Files Trim BuildSpec

- The kwtrimbuildspec script is used to ignore the 3rd party and unintended files\components.

Note: If file\folder exclusion is already existing for this program in Coverity, program team can reuse the same so that the respective files\components can be excluded from Klocwork analysis

- Once the above script execution is done, modified build spec will be generated as an output. Program team need to use the modified build spec for the Klockwork build analysis execution

- Ignore files\components list pattern:

Windows- `**\di.core.cmfb.sndctrl\**`
`**\di.core.OdoCtrl\src\file.c`

Linux- `**/codelinaro-aosp-caf/device/amlogic/yukawa**`
`**/codelinaro-aosp-caf/device/amlogic/yukawa-kernel/file.cpp`

- **Syntax:** `kwtrimbuildpec.exe -b <klocwork builds spec file> -p <list of ignore files or folders> -o <output of the klocwork builds spec name after ignore>.out`

Example Commands:

`kwtrimbuildpec.exe -b kwinject.out -p ignore_file_list.txt -o updatekwinject.out`

Klocwork Ignore Files Trim BuildSpec – Sample output

```
venka10@IND22BD2T3:~/NA_FORD_MY2025_P758$ ./kwtrimbuildpec -b vpinject.out -p ignore.txt -o ignore_vpinject.out
-----
kwtrimbuildpec.py      version: 1.2    created by: Perforce Software Inc
-----
[2024-03-22 16:58:41]: number of compiled files in build spec: 4074
Progress: |*****| 100.0% Complete
[2024-03-22 16:58:47]: number of compiled files excluded: 42
-----
FINISHED: SUCCESSFUL
Please reach out PCA team if any clarification
-----
```

3.SingleFileBuildspec

1. The SingleFileBuildspec This script is designed to trim the klocwork build spec compilation lines from a build specification based on path patterns provided, specifically for the Klocwork differential analysis.
 - Once the script execution is done, modified build spec will be generated as an output. Program team need to use the modified build spec for the further Klockwork analysis execution.
 - files\components list pattern:
Windows- `**\di.core.cmfb.sndctrl\source.java`
`**\di.core.OdoCtrl\src\file.java`

Linux- `**/codelinaro-aosp-caf/device/amlogic/source.java`
`**/codelinaro-aosp-caf/device/amlogic/yukawa-kernel/file.cpp`
 - **Syntax:** `./SingleFileBuildspec -b <klocwork builds spec file> -p <files list which are need to analysis> -o <output of the klocwork builds spec name after ignore>.out`

Example Commands:

`./SingleFileBuildspec -b kwinject.out -p file_list.txt -o [updatekwinject].out`

4.Klocwork AutoSconf – Identify huge issues(1/2)

- kwautosconf is used to identify huge defects reported in specific checkers for analyzed files in build.log
- This utility will block the commit and provide warning line(which are present in build.log) and create a sconf file where files with respective checkers have huge issues
- With this information, the integrators can share the generated sconf file with klocwork team or update their existing sconf file.
- To proceed, the analysis needs to be rerun with --force option or by deleting the existing kw_tables where the checkers-related defects are updated in filter.sconf will not be analyzed.
- Further to this commit can be performed.

AutoSconf – Identify huge issues(2/2)

- the latest script will create the sconf automatically, The Respective integrator needs to share the file with the klocwork team. The Klocwork team will add the sconf file to the respective project
- You need to place this script between kwbuildproject(analyze) command and kwadmin(load) command.
- **Syntax:** kwAutoSconf <kw_tables> auto

Example Commands:

```
kwAutoSconf C:\Users\mvenka10\Downloads\C519\gp_tables auto
```

The below screenshot shows using Autosconf Utility this huge issues warnings are populated as output. Based on this project team needs to update filter.sconf

(Huge issues were reported in Klocwork build.log for a checker MISRA.TOKEN.WRONGESC in a file static_tables.c)

Generated Autosconf outcome

```
C:\Volvo\Klocwork\Make_report_July_release\kwAutoSconf_V2.0\dist>kwAutoSconf.exe ..\build_RE-12 auto
CurrentVersion is 2.0
Exiting build.log: Files needing suppression:
File: **\algorithm Checkers: MISRA.GENFU.ASSOC
/data/rebuild/RE-Workspace/qssi-12/external/libcxx/include/algorithm
File: **\vector Checkers: MISRA.FUNC.DECL.AFTERUSE
/data/rebuild/RE-Workspace/qssi-12/external/libcxx/include/vector
File: **\DenseMap.h Checkers: MISRA.FUNC.DECL.AFTERUSE
/data/rebuild/RE-Workspace/qssi-12/external/llvm/include/llvm/ADT/DenseMap.h
File: **\STLExtras.h Checkers: MISRA.GENFU.ASSOC
/data/rebuild/RE-Workspace/qssi-12/external/llvm/include/llvm/ADT/STLExtras.h
```

Filter.sconf

```
{ "issue
_suppressions": [
  { "kind": "file",
    "name": "issue_codes \"static_tables.c\",
    \"\": [\"MISRA.TOKEN.WRONGESC\"],
    "filter": true
  }
}]}
```

5. Klocwork desktop analysis report XML to HTML and JSON/XML to CSV

- Once the Klocwork desktop command line analysis is done we can be able to generate the XML and the JSON results
- Using the latest scripts, we can be able to convert XML to CSV, XML to HTML, JSON to CSV
- In below we can see the sample outputs results

Klocwork Desktop Analysis Report

Report generated on: 07/05/2024 07:45 PM

ID	Origin	File	Line	Method	Code	Message	Status	Severity	Severity Level	Taxonomies	Reference
376732	System	C:\Venkatesh\Codebase\CL_Nissan_2018_L218_IC_Development\3rd.mcal.traveo.p2\plugins\Adc\lib\include\Adc_Types.h	84	Adc_Types.h	MISRA.DEFINE.BADEXP	Inappropriate macro expansion	Analyze	Review	4	CERT C Recommendations Levels MISRA Checkers	PRE02-C(L1), PRE10-C(L1)
376748	System	C:\Venkatesh\Codebase\CL_Nissan_2018_L218_IC_Development\3rd.mcal.traveo.p2\plugins\Adc\lib\include\Adc_Types.h	220	Adc_Types.h	PORTING.STORAGE.STRUCT	Byte position of elements in a structure could be changed depending on alignment and packing attributes.	Analyze	Review	4	C and C++ CERT C Rules Levels	DCL39-C(L3)
376749	System	C:\Venkatesh\Codebase\CL_Nissan_2018_L218_IC_Development\3rd.mcal.traveo.p2\plugins\Adc\lib\include\Adc_Types.h	285	Adc_Types.h	PORTING.STORAGE.STRUCT	Byte position of elements in a structure could be changed depending on alignment and packing attributes.	Analyze	Review	4	C and C++ CERT C Rules Levels	DCL39-C(L3)
421439	System	C:\Venkatesh\Codebase\CL_Nissan_2018_L218_IC_Development\3rd.mcal.traveo.d03\BSWA_Common\Platform_Types.h	89	Platform_Types.h	MISRA.DEFINE.BADEXP	Inappropriate macro expansion	Analyze	Review	4	CERT C Recommendations Levels MISRA Checkers	PRE02-C(L1), PRE10-C(L1)

ID	Status	Severity	Severity	Origin	Owner	File	Line	Method	Code	Message	Taxonomi	Reference
6047	Analyze	Review	4	System	unowned	/home/m	1501	SecM_Ver	MISRA.AR	Variable-I	MISRA C	18.8 (C99-2012 req.)
6048	Analyze	Review	4	System	unowned	/home/m	1528	SecM_Ver	MISRA.AR	Variable-I	MISRA C	18.8 (C99-2012 req.)
6049	Analyze	Review	4	System	unowned	/home/m	1136	SecM_Ver	MISRA.CA	Conversio	CWE for	704, 843
6050	Analyze	Review	4	System	unowned	/home/m	1244	SecM_Ver	MISRA.CA	Conversio	CWE for	704, 843
6051	Analyze	Review	4	System	unowned	/home/m	1336	SecM_Ver	MISRA.CA	Conversio	CWE for	704, 843
6052	Analyze	Review	4	System	unowned	/home/m	1378	SecM_Ver	MISRA.CA	Conversio	CWE for	704, 843
6075	Analyze	Review	4	System	unowned	/home/m	722	SecM_Upc	MISRA.ET	An expres	MISRA C	10.3 (C99-2012 req.)
6076	Analyze	Review	4	System	unowned	/home/m	622	SecM_Upc	MISRA.ET	The opera	MISRA C	10.4 (C99-2012 req.)
6077	Analyze	Review	4	System	unowned	/home/m	623	SecM_Upc	MISRA.ET	The opera	MISRA C	10.4 (C99-2012 req.)
6078	Analyze	Review	4	System	unowned	/home/m	708	SecM_Upc	MISRA.ET	The opera	MISRA C	10.4 (C99-2012 req.)
6079	Analyze	Review	4	System	unowned	/home/m	722	SecM_Upc	MISRA.ET	The opera	MISRA C	10.4 (C99-2012 req.)
6085	Analyze	Review	4	System	unowned	/home/m	1458	SecM_Init	MISRA.FU	Formal pa	MISRA C	2.7 (C99-2012 adv.)
6086	Analyze	Review	4	System	unowned	/home/m	1476	SecM_De	MISRA.FU	Formal pa	MISRA C	2.7 (C99-2012 adv.)
6087	Analyze	Review	4	System	unowned	/home/m	472	Sec_Verif	MISRA.PP	Function	MISRA C	8.13 (C99-2012 adv.)
6088	Analyze	Review	4	System	unowned	/home/m	806	Sec_Verif	MISRA.PP	Function	MISRA C	8.13 (C99-2012 adv.)
6089	Analyze	Review	4	System	unowned	/home/m	1228	Sec_Verif	MISRA.PP	Function	MISRA C	8.13 (C99-2012 adv.)

Klocwork Utilities Release Version



Utility Name	Latest Version	Release Date
kwcsvreport	V4.6	30 SEP 2024
kwautosconf	V2.0	9 JUL 2024
kwbuildsummary	V3.0	30 SEP 2024
kwtrimbuildpec	V2.0	30 APR 2024
kwoutputtocsv(Desktop)	V2.0	30 SEP 2024
Kwxmltohtml(Desktop)	V2.0	30 SEP 2024
kwsinglefilebuildspec(Desktop)	V1.1	30 APR 2024

- For any queries, please raise a Jira case if anyone faces any issues.
- Jira Link → <https://visteon.atlassian.net/jira/software/c/projects/PCA/boards/1171>
- Utility Location → https://visteon.sharepoint.com/sites/Wiki_Documents_Coverity/Shared%20Documents/Klocwork/Scripts/kw_utilities.7z?csf=1&web=1&e=rI0zX3
- Klocwork WIKI → https://visteon.sharepoint.com/sites/Wiki_Documents_Coverity/SitePages/Klocwork%20Wiki.aspx
 - IVI – Krishnan, Anuradha (A.) <akrish13@visteon.com>
 - DI – Krishnasamy, Vijaikumar (V.)<vkrish14@visteon.com>

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Thank You!!